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Task Description:	We have utilized 3D quantitative computed tomography to quantify sub-regional changes in vertebral and proximal femoral bone mineral density. In this project, we focus on the proximal femur, where we have found strikingly different patterns of cortical and trabecular bone loss in the eleven crew members with complete pre- and postflight measurements studies so far. This research has three primary aims. First to utilize patient specific finite element modeling and sectional bone strength estimates to relate the compartmental density measurements to changes in the biomechanical competence of the hip. Second, to develop a new image processing method to merge the individual CT images and finite element models to three-dimensionally visualize spatial patterns of change in bone mineral density, stress and strain in groups of individuals undergoing long-duration micro-gravity exposure. Third, to characterize the changes in the densities and volumes of the muscles adjoining the hip, and to relate these to the specific patterns of bone mineral loss. We believe that this research will provide insight into the physiological response of proximal femoral bone tissue to spaceflight and may result a more sensitive method for tracking the efficacy of countermeasures aimed at preserving skeletal strength. The overall objective of the task is to determine the extent to which hip strength is lost during spaceflight and whether it is recovered as the skeleton readapts to Earth gravity. The most important aspect of this task is the use of finite element analysis to analyze bone strength with respect to loads exerted by the hip in falls. The second objective is to combine the finite element analysis of the hip with inter-subject registration techniques so that it will be possible to visualize the overall response of the proximal femur at the organ level to changes in loading. The idea is to use the technique to identify regions of the hip that may be at highest risk for failure due to abnormal loading in falls or during excursion activities. The third objective is to develop the an approach for assessing muscles that are of key importance for protecting hip bone strength. If these muscles are identified, then it would be possible to reduce the time burden and injury risk associated with exercise programs by finding activities that are more efficient for protecting bone strength. This task represents the first use of bone strength rather than bone mass or density as an endpoint for skeletal health. In this sense, we will examine the loss and recovery of bone strength, develop a technique to better understand these changes, and develop an appaoch for evaluating the effect of muscle changes on bone strength.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All of the task objectives listed here benefit health on earth. The first objective allows us to understand the effect of disuse on bone strength as well as to work out the problems associated with using finite element analysis to perform serial studies. The second objective will provide a bone visualization technique which potentially will be just as effective for pharmacologic or disease studies as for spaceflight, and the muscle work in the third objective will allow us to potentially target muscle groups for exercise s designed to prevent or reduce postmenopausal bone loss in the hip.
	We have completed three project aims and are in the processing of preparing manuscripts on all aims. In the following, we list the task progress by each individual aim. Aim 1: Finite element analyses. We have installed finite element software developed by Dr Joyce Keyak in our laboratory at UCSF. The software, which creates biomechanical models of the hip from CT scans to estimate the whole bone strength of the proximal femur in loading conditions simulating normal ambulation and a fall to the side, has been applied to pre-flight, postflight and a subset of one year scans from the astronauts of increment 2-8. Finite element modeling of pre- and postflight scans and one year are complete for 16 subjects, and 5 one-year scans are available for data reduction as we are computing FEM-specific corrections for a scanner change that took place during the study. Our findings were presented as a podium presentation by co-investigator J Keyak at the 2007 Orthopaedic Research Society Meeting. After obtaining institutional review board approval and informed consent, pre- and post-flight CT scans were obtained of the hips of astronauts who participated in long-duration ISS missions (GE Hispeed Advantage; 3-mm-thick slices; 80 kVp; 280 mA; standard reconstruction). Mission length ranged from 4.3 to 6.5 months. Demographic data cannot be provided due to HIPAA regulations. A four-sample calcium hydroxyapatite (CHA) calibration phantom (Image Analysis, Columbia, KY) was placed under the subjects' hips during scanning and used to calibrate the CT scan data in terms of CHA concentration. From the CT scan data, a finite element (FE) model of each subject's left proximal femur was generated using methods described previously. These models employed patient-specific geometries and material properties. Ash density (rho ash, g/cm3) of each element was computed from the calibrated CT scan data (rho QCT, g/cm3), rho ash=0.0633 + 0.887 rho QCT. Mechanical properties were calculated from as densities.These properties were calibrated to result in accurate predictions of FFE.4 The models were analyzed under loading simulating single-limb stance2, and impact from a fall onto the greater trochanter. For stance loading, force was applied to the femoral head at 20° to the shaft axis, within the coronal plane. For fall loading, force was applied to the femoral head at 60° to the shaft axis and 25° to the coronal plane, as the opposing surface of the greater trochanter was restrained in the opposite direction. FE-computed hip strength (FFE) in each loading condition was determined. A paired t-test was used to evaluate differences between pre- and post-flight values of FFE. RESULTS: For stance loading, the average proximal femoral strength decreased significantly ($p<0.001$) from 13,200 N (SD=2300 N) pre-flight to 11,200 N (SD=2400 N) post-flight . The average rate of loss of hip strength for stance loading was 2.6%/mo (SD=1.4%). For fall loading, average hip strength decreased significantly ($p=0.003$) from 2,580 N (SD=560 N) pre-flight to 2,280 N (SD=590 N) post-flight . The average rate of strength loss for fall loading was 1.9%/mo (SD=1.4%). The preliminary one year FEM data show that for 4/5 subjects, strength is recovered in the stance loading condition, but for 4/5 subjects strength is not recovered in the fall loading condition. DISCUSSION: This study has shown that astronauts exposed to microgravity for 4.3 to 6.5 months lose hip strength at an average rate of 2.6%/mo under single-limb stance loading and 1.9%/mo under fall loading. Previous DXA measurements of astronauts returning from the ISS indicated that aBMD decreased by 1.4%/mo in the neck and 1.5%/mo in the total femur.1 Thus, estimates of bone strength loss that are based on DXA aBMD alone will understate the reduction in hip strength.The receovery data available thus far seem to be consistent with our finding in our 2006

Task Progress:

bone recovery paper, that shows that the recovery of bone mass and areal density is nearly complete at one year, but that volumetric bone density, particularly the density of trabecular bone does not recover. This seems to be driven by periosteal expansion, which increases bone volume and mass, while the recovery of volumetric bone density lags. The larger bone may be optimized for the loads incurred in normal usage but may still be at risk for fracture in pathologic conditions such as falls. Thus, recovery of bone mass may not be sufficient to gauge the recovery of bone after spaceflight.

Aim 2: Intersubject registration

This aim is complete. We have obtained the Montreal Neurological Institute Animal intersubject image registration program, installed it in our laboratory, and have applied it to successfully co-register CT scans of the hip from 16 subjects with pre and postflight image data. We have carried out voxel by voxel statistical comparisons of those models for the pre- and postflight time points, and have demonstrated a focal pattern of accelerated bone loss in the compressive trabecular band in the femoral head region. In June 2007, we published this research in the journal BONE (W Li et al, BONE epub June 2007, see bibliography)

Aim 3: Quantitation of muscle cross-sectional area and attenuation coefficient from Hip CT scans.

We have completed the technical development for this task and are in the process of analyzing hip muscle variables in our cohort and in studies of women with and without hip fracture.

Acquisition of volumetric CT scans of the hip: Analysis of proximal femoral musculature is carried out using volumetric CT scans optimized to analyze the density and structure of the proximal femur. In these studies, the patient lies supine on the imaging table, and an anteroposterior scoutview is acquired which spans the iliac crest to the mid-thigh. The operator then defines a helical scan volume between the superior limit of the left femoral head and an inferior limit 3-5 mm inferior to the lesser trochanter. The muscle analyses can carried out on images obtained with 120/80 kVp, 150 mAs, 3 mm or 1-mm section thickness (pitch=1), 48 cm field of view, reconstructed with a soft tissue filter. Images in all studies were obtained with a QCT calibration phantom (Image Analysis, Columbia, KY) positioned below the patient's hips.

Analysis of images for muscle cross-sectional area and mean attenuation coefficient: Based on previous studies of gait, balance and muscle function, we have selected four groups of muscles to quantify which are well-visualized within the field of view of the proximal femur QCT scan: these include the hip extensors (gluteus maximus), hip abductors (gluteus medius, gluteus minimus and piriformis), hip adductors (adductor magnus, adductor brevis, adductor longus, quadratus femoris, quadratus externus, pectineus) and hip flexors (rectus femoris and sartorius). These groups are easily distinguishable on almost all of the cross-sections in the field of view in CT scans of the proximal femur. Because the cross-sectional areas of these muscle groups vary differently as a function of slice location within the QCT scans, we developed a method to select individual cross-sections on which to quantify the areas and mean attenuation coefficients of these muscle groups. This involved two steps. First we examined annotated optical and CT images in the Visible Human data set to determine the locations in the CT scan field of view where the muscle groups are of maximal size and are optimally distinguished from the surrounding tissue, and second we developed a protocol for employing boney landmarks to calculate the locations of these cross-sections relative to the hip anatomy, so that the muscle measurements could be compared between individuals and reproduced within an individual over time. To select the slice location, a mid-coronal reformation of the image is employed to display the proximal femoral anatomy. The user marks the slice position of the superior aspect of the femoral head and the inferior aspect of the lesser trochanter to generate a metric by which the individual muscle cross-sections can be reliably located. SH is the slice position corresponding to the superior aspect of the left femoral head. D is the distance between SH and the inferior aspect of the lesser trochanter, which defines the inferior limit of the proximal femoral analysis. The cross-sections are chosen as the slice numbers corresponding to the nearest integer to the calculations. Thus three cross-sections are selected for display, one for the extensors, one for the abductors, and a third cross-section containing the adductors and flexors. Each slice is then displayed in an image window, and the user draws the boundary around the fasceal plane of the corresponding muscle group. Once the drawings have been completed, the program processes each cross-section according to the following steps:

1) The program finds the water equivalent cell of the calibration phantom. Based on the mean and standard deviation of the HU in the water equivalent cell, the program calculates HU intervals to be used for calculations of muscle and fat areas within defined tissue boundaries.

2) The program generates the outer body contour by performing a region growing operation, searching for the skin-air boundary on each section from a user-placed seed point inside one of the muscle groups. Following that, the program performs a region growing, starting from the same seed point, finding the cortical bone/muscle boundary. The region within the bone boundary is then excluded from subsequent soft tissue calculations.

3) The program then starts at the body contour, and grows inward, searching for the fat/muscle boundary. This provides mask of lean tissue.

Findings:

Astronaut study: We have applied this methodology to pre- and post-flight scans of our astronaut data, including 16 subjects with pre- and postflight QCT scans. Comparison of pre- and post-flight scans showed 1-5% losses in total lean area and in the areas of the flexor, abductor, adductor and extensor muscle groups, with only a 5% loss in the adductor area being statistically significant (p=0.02) Small losses may reflect effect of inflight and postflight exercise protocols and the acquisition of CT data over 2-4 weeks post-return, allowing for some recovery of muscle area.

Hip fracture study: This work has been accepted as a podium presentation at the 2007 American Society of Bone and Mineral Research Meeting. To test the hypothesis that hip muscle size and attenuation are associated with hip fracture, we compared CT scans of the contralateral hip in 45 women (mean age 74.71±5.94) imaged within 48 hours of their hip fractures to 66 age-matched control subjects (mean age 70.70±4.66). A volumetric CT scan encompassing the proximal part of the femur was analyzed for total percentage of fat tissue in the in the CT field of view and for the CSA and lean tissue Hounsfield Unit (HU) values of the hip extensor, abductor, adductor and flexor muscles. The CT scan was also analyzed to compute areal bone mineral density (aBMD) of the hip bone. Logistic regression analyses were employed to determine differences between fracture subjects and controls in body composition measures after adjustment for age, height, and BMI. Receiver-operator curve (ROC) analyses were employed to determine if combinations of aBMD and body composition were better discriminators of hip fracture than aBMD alone.

	We found that subjects with hip fractures were on average 5 years older (p<0.001) had 8% lower BMI (p<0.05), 8% lower percentage fat (p<0.0001) and roughly 20% lower extensor and adductor CSA values than controls (p<0.0001). Despite having lower total adiposity, they had higher muscle adiposity as indicated by lower lean tissue attenuation in all four muscle groups, with statistically significant differences in the adductor, abductor and flexor groups. After adjustment for age and BMI, the fracture subjects had lower total % fat, smaller extensor and adductor CSA values, and greater adipose infiltration (lower attenuation) in the adductor (0.001 (less than) p (less than) 0.05). ROC analyses showed that models combining percent fat, muscle CSA and muscle attenuation values with aBMD had higher AUC values than models containing only aBMD (0.97 vs 0.88, p<0.001). We conclude that reduced total adiposity and increased hip muscle adiposity appear to be independent correlates of hip fracture, and combination of skeletal and body composition variables appears to enhance diagnostic efficacy compared to skeletal variables alone. We found that subjects with hip fractures were on average 5 years older (p<0.001) had 8% lower BMI (p<0.05), 8% lower percentage fat (p<0.0001) and roughly 20% lower extensor and adductor CSA values than controls (p<0.0001). Despite having lower total adiposity, they had higher muscle adiposity as indicated by lower lean tissue attenuation in all four muscle groups, with statistically significant differences in the adductor, abductor and flexor groups (0.00001(less than) p (less than) 0.02). After adjustment for age and BMI, the fracture subjects had lower total % fat, smaller extensor and adductor CSA values, and greater adipose infiltration (lower attenuation) in the adductor (0.001(less than) p (less than) 0.05). ROC analyses showed that models combining percent fat, muscle CSA and muscle attenuation values with aBMD had higher AUC values than models containing only aBMD (0.97 vs 0.88, p<0.001). We conclude that reduced total adiposity and increased hip muscle adiposity appear to be independent correlates of hip fracture, and combination of skeletal and body composition variables appears to enhance diagnostic efficacy compared to skeletal variables alone.
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